INORGANIC CHEMISTRY CUMULATIVE EXAM

January 15,1994

One of the attractive features of transition metal chemistry is the variety of colors exhibited by metal complexes. As we all know these colors are for the most part determined by the nature of electronic transitions that can take place in these complexes. Absorption bands resulting from these electronic transitions are characterized by three features: 1) the energy, 2) the intensity or band height and 3) the band width. Using realistic examples of octahedral and tetrahedral transition metal complexes explain how each of these features is determined by the geometric and electronic structure of the complexes. The following points should be included:

1. Differences between octahedral and tetrahedral complexes

2. Laporte forbidden transitions

3. Spin forbidden transitions

4. MLCT and LMCT transitions

5. Difference between electronic states and electronic configurations